Real-Time Emotion Recognition System

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Introduction

- Emotion recognition is an application of AI and Computer Vision.
- Helps machines understand human emotions through facial expressions.
- Used in healthcare, security, and human-computer interaction.
- This system detects emotions in real-time using webcam.

Problem Statement

- Machines cannot naturally understand human emotions.
- Need for automated emotion detection systems.
- Build a real-time accurate and efficient system.

Objectives

- Develop CNN-based emotion classification model.
- Train model using facial expression dataset.
- Integrate with real-time webcam.
- Achieve accurate and fast predictions.

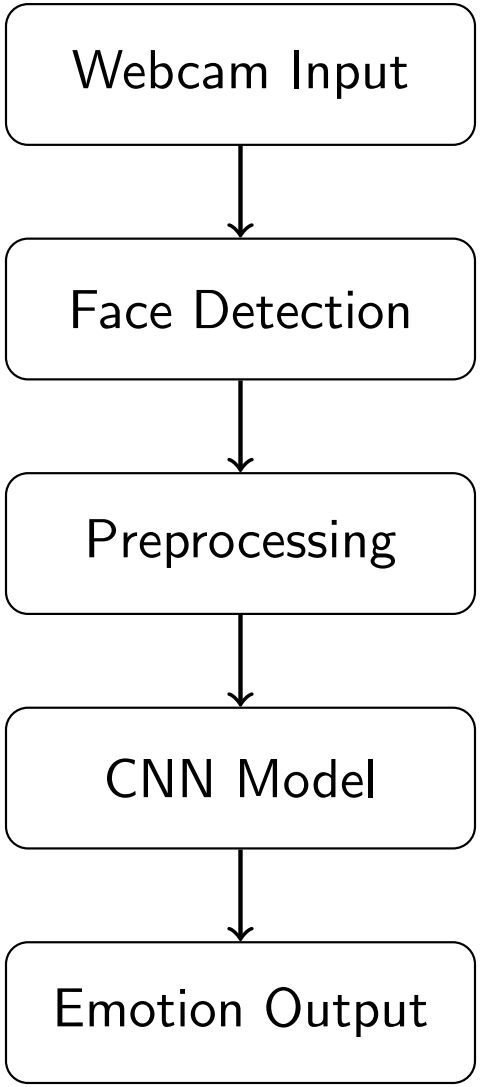
Scope of the Work

- Detect emotions like Happy, Sad, Angry, Neutral, Fear, Surprise.
- Real-time processing.
- Extendable to web and mobile applications.
- Useful in mental health monitoring.

Technologies Used

- Python
- TensorFlow / Keras
- OpenCV
- NumPy
- CNN (Deep Learning)

System Design



Methodology

1. Dataset collection
2. Data preprocessing
3. Model training
4. Evaluation
5. Real-time deployment

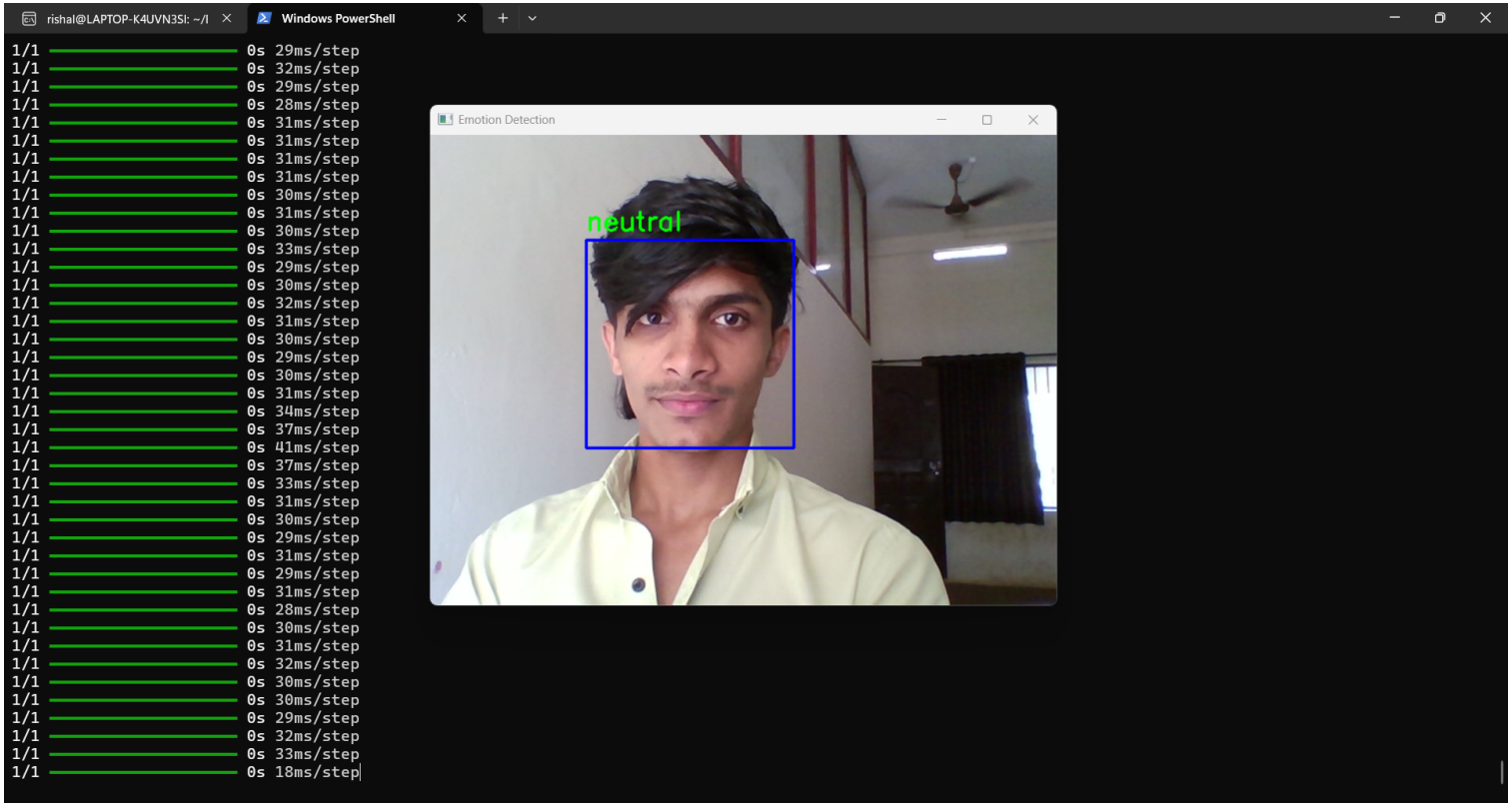
Implementation

- Used FER dataset
- Built CNN with multiple layers
- Trained model for classification
- Saved model in .keras format
- Integrated with OpenCV

Output Screenshots

- Real-time emotion detection
- Face bounding box
- Emotion labels displayed

Output 1


rishal@LAPTOP-K4UVN3SI: ~/I
Windows PowerShell
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1/1 0s 32ms/step
1/1 0s 29ms/step
1/1 0s 28ms/step
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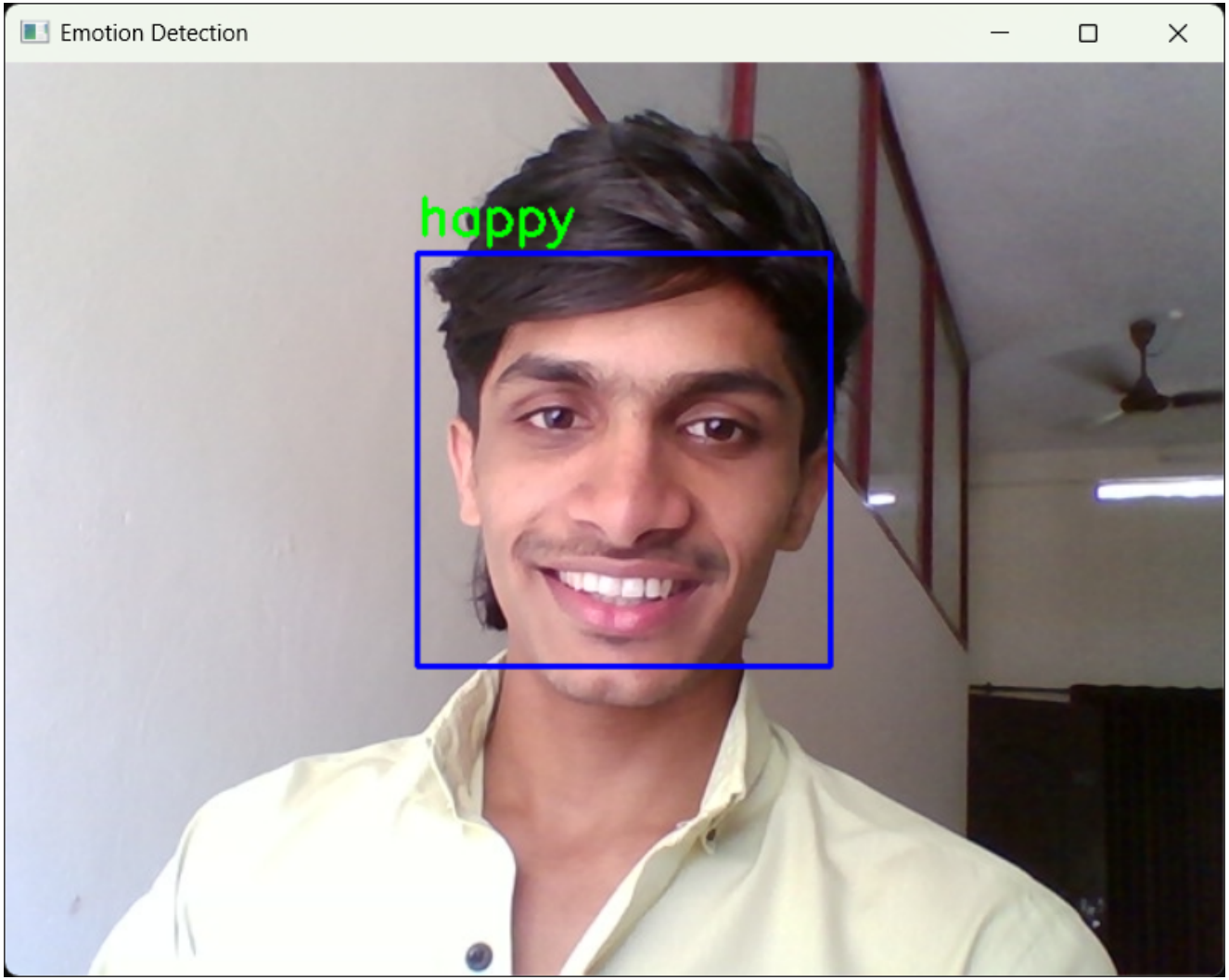
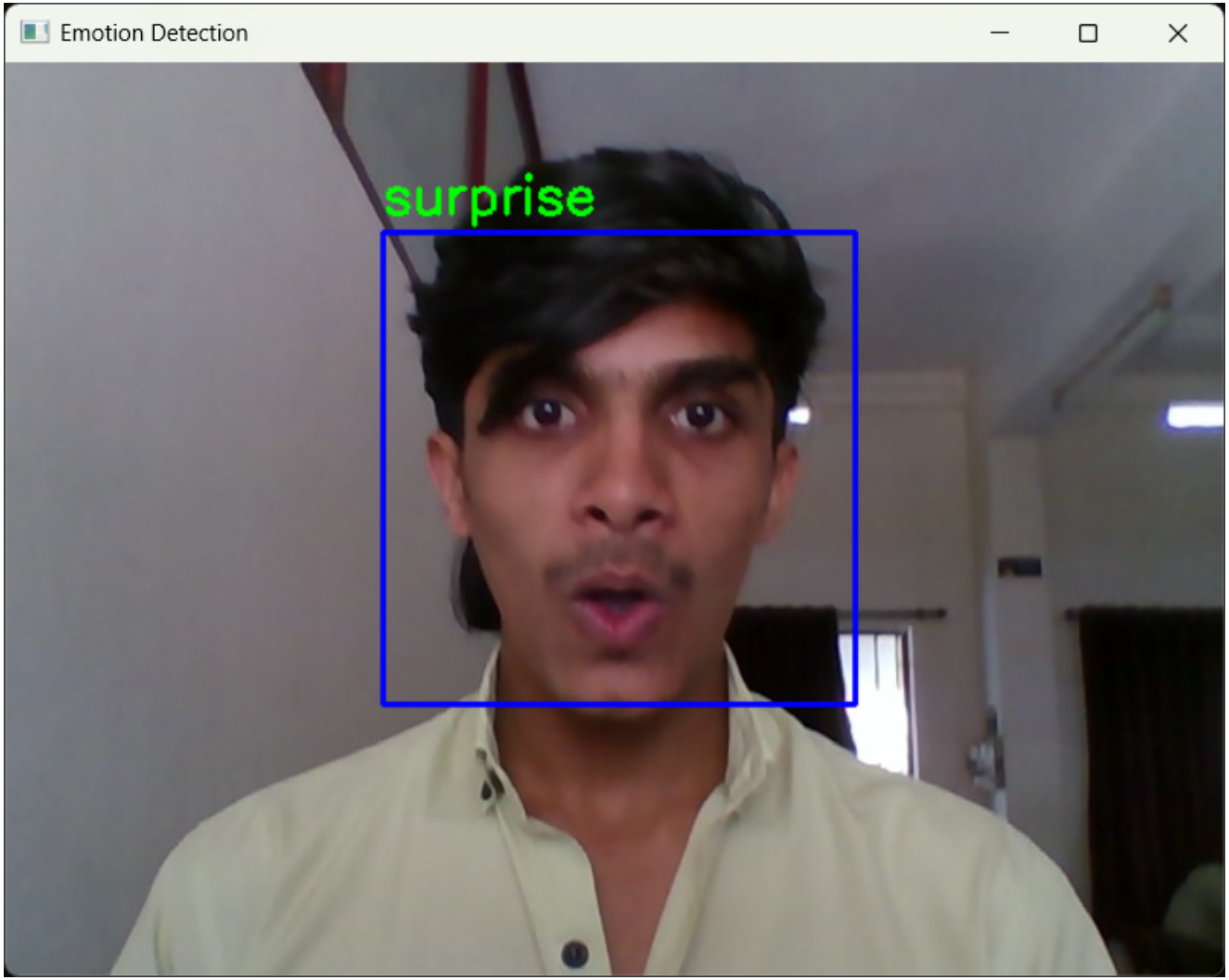
Emotion Detection
neutral

Output 2



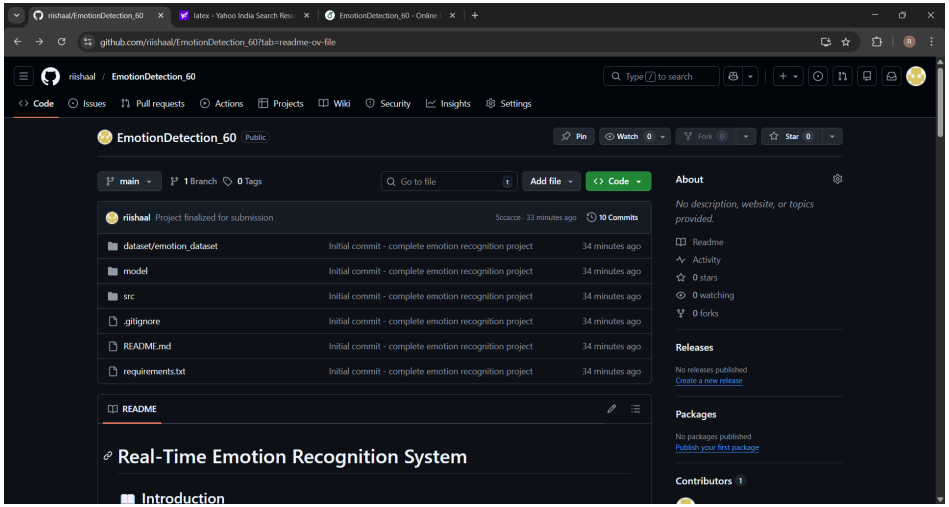
Output 3



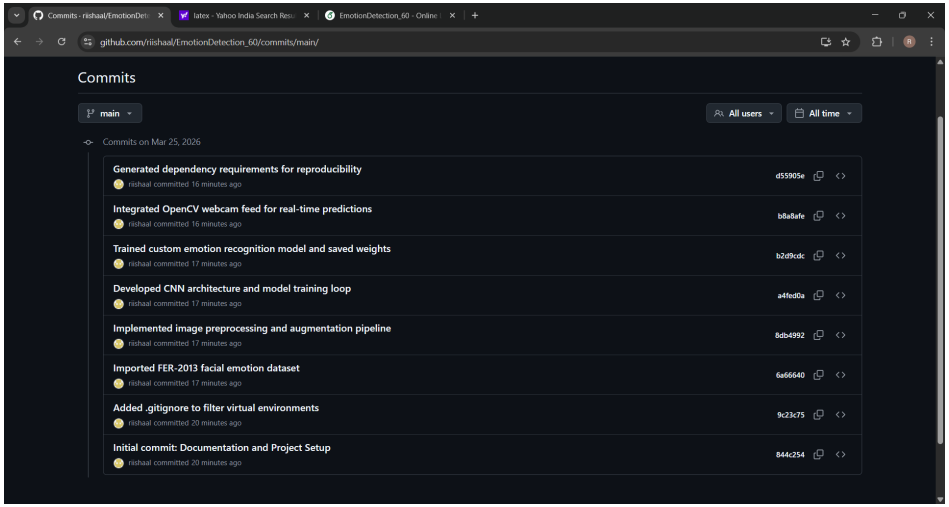
GitHub Overview

GitHub Link:

https://github.com/riishaal/EmotionDetection$_6$0



Repository Structure



Commit History

Challenges Faced

- Dataset imbalance
- Accuracy improvement
- Real-time processing
- Platform compatibility issues

Results and Conclusion

- Successfully implemented real-time system
- Achieved 55% accuracy
- Efficient on CPU
- Demonstrates AI application

Future Scope

- Improve accuracy using advanced models
- Deploy as mobile/web app
- Use transfer learning
- Add analytics

References

- TensorFlow Documentation
  https://www.tensorflow.org/
- OpenCV Documentation
  https://opencv.org/
- FER-2013 Dataset (Kaggle)
  https://www.kaggle.com/datasets/msambare/fer2013
- Keras Documentation
  https://keras.io/
- Facial Expression Recognition using CNN
  IEEE Paper / Google Scholar